

## Result of running mvn package:

```
[INFO] Running CSE464ProjectP1.GraphTest
[INFO] Tests run: 10, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.026 s -- in CSE464ProjectP1.GraphTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 10, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ CSE464_2026_lseekalr ---
[INFO] Building jar: /Users/lisharamamurthy/CSE464/CSE464-P1/target/CSE464_2026_lseekalr-1.0-SNAPSHOT.jar
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 0.987 s
[INFO] Finished at: 2026-04-03T16:35:07-07:00
[INFO]
```

## Github Actions:

[https://github.com/lisha080/CSE464\\_2026\\_lseekalr/actions/runs/23967233514](https://github.com/lisha080/CSE464_2026_lseekalr/actions/runs/23967233514)

The screenshot shows the GitHub Actions interface for a workflow named 'Update .gitignore #12'. The workflow is currently in a 'build' state, which succeeded 5 minutes ago. The left sidebar shows the workflow file and a list of jobs, with 'build' selected. The main area displays the 'Annotations' for the 'build' job, showing a list of steps and their durations:

| Step                         | Duration |
|------------------------------|----------|
| Set up job                   | 1s       |
| Run actions/checkout@v3      | 1s       |
| Set up JDK                   | 4s       |
| Install Graphviz             | 15s      |
| Build with Maven             | 18s      |
| Post Set up JDK              | 8s       |
| Post Run actions/checkout@v3 | 8s       |
| Complete job                 | 8s       |

\*\*\*early commits failed with CI due to Java version issues and the later ones failed due to graphviz spelling mistakes

## Github Commits:

### Main:

Feature 5 Remove node, Feature 6 Remove nodes, Feature 7 Remove edge:

[https://github.com/lisha080/CSE464\\_2026\\_lseekalr/commit/2beb5b052784fd29741e53a96228ca95a8c468e8](https://github.com/lisha080/CSE464_2026_lseekalr/commit/2beb5b052784fd29741e53a96228ca95a8c468e8)

Merge Conflict:

- [https://github.com/lisha080/CSE464\\_2026\\_Iseekalr/commit/63a2a0297b35add5600d33f3b06e20c6038f9bcf](https://github.com/lisha080/CSE464_2026_Iseekalr/commit/63a2a0297b35add5600d33f3b06e20c6038f9bcf)
- [https://github.com/lisha080/CSE464\\_2026\\_Iseekalr/commit/5aeff40c45ddef784c20fa9ab51c0484f9007757](https://github.com/lisha080/CSE464_2026_Iseekalr/commit/5aeff40c45ddef784c20fa9ab51c0484f9007757)

Continuous Integration:

[https://github.com/lisha080/CSE464\\_2026\\_Iseekalr/commit/89c67515527a24d8accc9140882aa7792c09562f](https://github.com/lisha080/CSE464_2026_Iseekalr/commit/89c67515527a24d8accc9140882aa7792c09562f)

### **BFS:**

Feature 8 BFS search:

[https://github.com/lisha080/CSE464\\_2026\\_Iseekalr/commit/e750590a5cd79e23ff7d8190d87be6e284575ba7](https://github.com/lisha080/CSE464_2026_Iseekalr/commit/e750590a5cd79e23ff7d8190d87be6e284575ba7)

### **DFS:**

Feature 9 DFS search:

[https://github.com/lisha080/CSE464\\_2026\\_Iseekalr/commit/ffef9f1eeadba093736c944908cf0f04b5b64047](https://github.com/lisha080/CSE464_2026_Iseekalr/commit/ffef9f1eeadba093736c944908cf0f04b5b64047)

### **Branches links:**

bfs: [https://github.com/lisha080/CSE464\\_2026\\_Iseekalr/tree/bfs](https://github.com/lisha080/CSE464_2026_Iseekalr/tree/bfs)

dfs: [https://github.com/lisha080/CSE464\\_2026\\_Iseekalr/tree/dfs](https://github.com/lisha080/CSE464_2026_Iseekalr/tree/dfs)

## Features with input + output

### Feature 1: Parse Graph

```
// Feature 1 - Parse graph
Graph graph = new Graph();
graph.parseGraph( filepath: "input.dot");
System.out.println("=== Feature 1: Parse Graph ===");
System.out.println(graph);
```

```
=== Feature 1: Parse Graph ===
Number of nodes: 8
Node labels:
a
b
c
d
e
f
g
h
Number of edges: 9
Edges:
a -> b
b -> c
c -> d
d -> a
a -> e
e -> f
e -> g
f -> h
g -> h
```

### Feature 2: Add nodes

```
// Feature 2 - Add nodes
graph.addNode( label: "x");
graph.addNodes(new String[]{"y","z"});
System.out.println("=== Feature 2: Add Nodes ===");
System.out.println(graph);
```

```
=== Feature 2: Add Nodes ===  
Number of nodes: 11  
Node labels:  
a  
b  
c  
d  
e  
f  
g  
h  
x  
y  
z  
Number of edges: 9  
Edges:  
a -> b  
b -> c  
c -> d  
d -> a  
a -> e  
e -> f  
e -> g  
f -> h  
g -> h
```

### Feature 3: Add edges

```
// Feature 3 - Add edges  
graph.addEdge( srcLabel: "x", dstLabel: "a");  
graph.addEdge( srcLabel: "a", dstLabel: "b");  
System.out.println("=== Feature 3: Add Edges ===");  
System.out.println(graph);
```

```
=== Feature 3: Add Edges ===  
Number of nodes: 11  
Node labels:  
a  
b  
c  
d  
e  
f  
g  
h  
x  
y  
z  
Number of edges: 10  
Edges:  
a -> b  
b -> c  
c -> d  
d -> a  
a -> e  
e -> f  
e -> g  
f -> h  
g -> h  
x -> a
```

#### Feature 4: Output PNG and DOT files

```
// Feature 4 - Output DOT and PNG
graph.outputDOTGraph( path: "output.dot");
graph.outputGraphics( path: "output.png", format: "png");
System.out.println("=== Feature 4: Output DOT and PNG ===");
System.out.println(graph);
```

```
=== Feature 4: Output DOT and PNG ===
Number of nodes: 11
Node labels:
a
b
c
d
e
f
g
h
x
y
z
Number of edges: 10
Edges:
a -> b
b -> c
c -> d
d -> a
a -> e
e -> f
e -> g
f -> h
g -> h
x -> a
```

#### Feature 5: Remove node

```
// Feature 5 - Remove node
System.out.println("=== Feature 5: Remove Node ===");
graph.removeNode( label: "x");
System.out.println("After removing node 'x':");
System.out.println(graph);
```

```
=== Feature 5: Remove Node ===  
After removing node 'x':  
Number of nodes: 10  
Node labels:  
a  
b  
c  
d  
e  
f  
g  
h  
y  
z  
Number of edges: 9  
Edges:  
a -> b  
b -> c  
c -> d  
d -> a  
a -> e  
e -> f  
e -> g  
f -> h  
g -> h
```

#### Feature 6: Remove nodes

```
// Feature 6 - Remove nodes  
System.out.println("=== Feature 6: Remove Nodes ===");  
graph.removeNodes(new String[]{"y","z"});  
System.out.println("After removing nodes 'y' and 'z':");  
System.out.println(graph);
```

```
=== Feature 6: Remove Nodes ===
After removing nodes 'y' and 'z':
Number of nodes: 8
Node labels:
a
b
c
d
e
f
g
h
Number of edges: 9
Edges:
a -> b
b -> c
c -> d
d -> a
a -> e
e -> f
e -> g
f -> h
g -> h
```

## Feature 7: Remove Edge

```
// Feature 7 - Remove edge
System.out.println("=== Feature 7: Remove Edge ===");
graph.removeEdge( srcLabel: "a", dstLabel: "b");
System.out.println("After removing edge a->b:");
System.out.println(graph);
```

```
=== Feature 7: Remove Edge ===
After removing edge a->b:
Number of nodes: 8
Node labels:
a
b
c
d
e
f
g
h
Number of edges: 8
Edges:
b -> c
c -> d
d -> a
a -> e
e -> f
e -> g
f -> h
g -> h
```

#### Feature 8: BFS Search

```
// Feature 8 - BFS search
System.out.println("=== Feature 8: BFS Graph Search ===");
Graph graph2 = new Graph();
graph2.parseGraph( filepath: "input.dot");
Path bfsPath = graph2.graphSearch( src: "a", dst: "h", Graph.Algorithm.BFS);
System.out.println("BFS path from a to h: " + bfsPath);
```

```
=== Feature 8: BFS Graph Search ===
BFS path from a to h: a -> e -> f -> h
```

#### Feature 9: DFS Search

```
// Feature 9 - DFS search
System.out.println("=== Feature 9: DFS Graph Search ===");
Path dfsPath = graph2.graphSearch( src: "a", dst: "h", Graph.Algorithm.DFS);
System.out.println("DFS path from a to h: " + dfsPath);
```

```
=== Feature 9: DFS Graph Search ===
DFS path from a to h: a -> e -> f -> h
```